MILITARY SPECIFICATION

CARTRIDGE, INITIATOR, XM201 ASSEMBLY

1. SCOPE

1.1 Scope.- This specification covers Cartridge, Initiator, XM201. Assembly will ultimately be used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-H-25579	- Hose, Assembly, Tetrafluoroethylene, High Temperature, Power Plant, Aircraft

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

U.S. Army Munitions Command

D10535460	- Cartridge, Initiator, XM201 Assembly
C8837884	- Carton, Packing, Ammunition for Cartridge, Initiator M91 and XM201
C8837885	- Box, Packing, Ammunition for Cartridge, Initiator M91 and XM201
IEL 10535460	- Index to Inspection Equipment List for Cartridge, Initiator XM201 Assembly

FSC 1375

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement function shall be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, propellant or derangement of any component.

3.4 Primer sensitivity.- The primer shall fire when drop tested at an energy level of 16 inch ounces in a sectioned cartridge, void of propellant charge.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces regardless of configuration.

3.6 Functional.

3.6.1 The cartridge shall not fail to fire.

3.6.2 Ballistics.- The cartridge when tested in an XM98 Initiator, shall produce the pressures and ignition delay at the temperatures specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, size 4, in accordance with MIL-H-25579 and a pressure chamber with a volume of 0.062 cubic inches. (See Fig. 1)

TABLE I

Temperature °F (Degrees Fahrenheit)	Pressure Pounds per square inch (PSI) Min. (See 6.1)	Max.	Ignition Delay Milliseconds, Max. (See 6.2)
-65° to -70°	2500	4000	65
65° to 75°	2600	4000	65
200° to 205°	2750	4100	65

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3.7 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings and the following:

3.7.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, perforations, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 80 cartridge assemblies and 10 sets of unassembled parts. The sample shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency a lot shall be 600 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Acceptance Numbers Major	Minor
1-300	100% examination	0	0
301-1300	110	1	2
1301-3200	150	2	3

No. of Defects in Class	Total Major Defects Allowable
2-3	1
4-7	2
8-19	3

No. of Defects in Class	Total Minor Defects Allowable
2-3	2
4-7	3
8-16	4

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4.3.2.1 Primer, Percussion, M29A1, Dwg. C8798312.

	Defect	Method of Inspection
CRITICAL		
1.	Any part missing	X-ray 1/
2.	Improper assembly	X-ray 1/
3.	Mixed primers	Visual
4.	Primer pellet broken, cracked or dislodged	Visual
MAJOR		
101.	Burr	Visual
102.	Foreign matter	Visual
MINOR		
	None defined	

1/ Primer X-ray.- Prior to assembly into a cartridge, the primer shall be x-rayed with the sides of the cup perpendicular to the rays to determine compliance with the requirements of the applicable drawing. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.2 Cartridge, Initiator, XM201 Assembly, Dwg. D10535460.

	Defect	Method of Inspection
CRITICAL		
1.	Any part missing	X-ray and Visual 1/
2.	Improper Assembly	X-ray and Visual 1/
3.	Mixed Ammunition	Visual
4.	Crimp missing or incorrect	Visual
5.	Marking missing, incorrect or illegible	Visual
6.	Metal defective	Visual
MAJOR		
101.	Concentricity of head with body	Gage
102.	Diameter of body, max.	Gage
103.	Diameter of head, max.	Gage
104.	Diameter inside head, min.	Gage
105.	Depth to surface "S"	Gage
106.	Location of datum diameter, max.	Gage
107.	Firing plug above flush with retaining cup	Gage
108.	Retaining cup above flush with surface "S"	Gage

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Defect	Method of Inspection
MAJOR, Cont.	
109. Protective finish inadequate or defective	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Dent, fold, wrinkle, buckle or bulge	Visual
203. Draw scratch	Visual
204. Varnish missing or inadequate	Visual
205. Surface finish improper	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.3 Carton, Packing, Ammunition for Cartridge, Initiator, M91 and XM201, Dwg. C8837884.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual and Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated carton	Visual
206. Marking missing, incorrect or illegible	Visual
207. Carton sealing defective, improper or missing	Visual

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4.3.2.4 Box, Packing, Ammunition for Cartridge, Initiator, M91 and XM201, Dwg. C8837885.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual and Visual
203. Contents of carton exposed	Visual
204. Filler or separator missing	Visual
205. Carton damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Sealing improper, missing or insecure	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4

4.3.3.1 Vibration.- Twelve of the 30 cartridge assemblies of the initial production sample shall be vibrated. The 12 cartridge assemblies shall then be x-rayed in accordance with note 1/ of 4.3.2.2. Failure of any cartridge assembly to comply with 3.3 shall be cause for rejection of the lot. Any change in the primer lot in subsequent production shall be reason to select an additional sample of 12 cartridge assemblies and subject it to the vibration test.

4.3.3.2 Primer sensitivity.- A sample of 50 assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for sensitivity. Failure of any cartridge assembly to comply with the requirement of 3.4 shall be cause for rejection of the lot.

4.3.3.3 Air test.- Each cartridge shall be subjected to the specified pressure differential. Failure of any cartridge to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.4 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing.

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4.3.3.5 Functional.- The following test shall be performed by the Government:

4.3.3.5.1 Ballistics.- Thirty units from each lot shall be grouped, temperature conditioned as specified in Table IV and 4.4.5.1.1 and tested in accordance with 4.4.5.1. Failure of any cartridge to comply with the requirements of 3.6 shall be cause for rejection of the lot.

TABLE IV

Temperature	Production Lot
-65° to -70°F	10
65° to 75°F	10
200° to 205°	10

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10535460.

4.4 Test Methods and Procedures

4.4.1 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII. The cartridge assemblies used for this test shall be utilized for the ballistic test as specified in 4.4.5.1.

4.4.2 Primer sensitivity.- Remove and discard all powder charges from the test cartridge by sectioning the case around its periphery 1/2" maximum above the base. Drop test sectioned cartridge at 8.165 minus .030 inches, using a steel ball 15/16" diameter, weighing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine, with a cartridge holder approved by the procuring activity.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL 10535460.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.5 Functional

4.4.5.1 Ballistics.- The cartridge shall be loaded into an acceptable XM98 Initiator, divided into three equal groups and temperature conditioned in accordance with Table IV. The Initiator shall have end caps to cover the gasport during the conditioning period. The cartridge shall be tested within 15 minutes after removal from the conditioning chamber. (See Fig. 1) When this time is exceeded, the Initiator shall be reconditioned at the original temperature.

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4.4.5.1.1 Temperature conditioning.- A standard XM98 Initiator loaded with a dummy XM201 Cartridge containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8837884 and C8837885.

5.2 Data cards.- Data card information shall be in accordance with MIL-A-2550.

6. NOTES

6.1 Advisory propellant loading data:

Pellet, 2D, Source: U. S. Flare Division of Atlantic Research Corp., Saugus, Calif.	13 pieces

6.2 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Custodian:

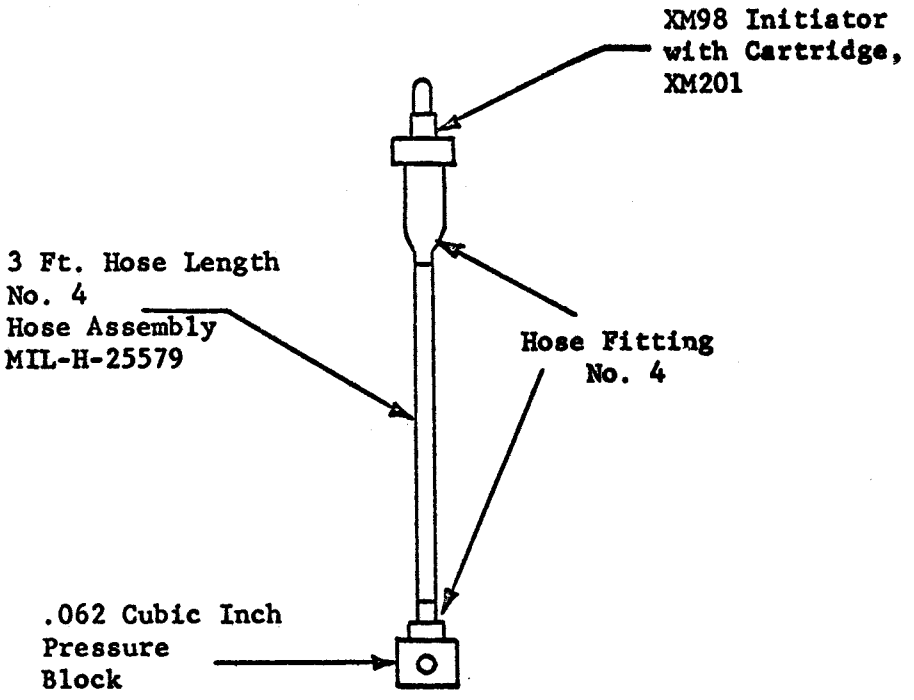
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Preparing activity:

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PERFORMANCE TEST LAYOUT SKETCH

TOP VIEW

Figure 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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